

Trading Strategies Reference

Purpose

This document is the single consolidated trading playbook for this project. It synthesizes the four source references:

- Docs/Statistical Arbitrage by Cointegration.pdf
- Docs/Volatility Trading.pdf
- Docs/Gamma Scalping.pdf
- Docs/Vega Scalping.pdf

Its goal is to turn the source material into an execution-ready reference for a retail-oriented, AI-assisted trading system. It keeps the original institutional logic, but adds practical controls for:

- retail capital limits
- option liquidity constraints
- transaction costs and slippage
- supervised execution
- model-risk handling
- scenario-based trade management

This document is a trading-methodology reference for the project, not a promise of profitability and not a substitute for broker, legal, tax, or compliance review.

Project Execution Assumptions

This project is designed around a supervised-first operating model. The practical implication for this document is:

- discretionary entries should be reviewed and approved by the operator before broker submission
- mechanical hedges, stop logic, and risk-reduction actions can be automated once a position is live
- promotion from supervised to higher autonomy should happen only after validated paper trading, realistic cost modeling, and stable live-like behavior

How To Use This Document

Use this reference in five steps:

1. Identify the strategy family that matches the market condition.
2. Validate the shared prerequisites in Common Execution Framework.
3. Follow the strategy-specific Entry, Sizing, Management, and Exit rules.
4. Check the scenario section for the situation you are actually in.

5. Do not execute if the setup fails any kill condition, even if the signal is attractive.

Common Execution Framework

Shared Principles

All four source strategies rely on the same institutional ideas:

- trade relationships or volatility instead of pure direction whenever possible
- remove unwanted risk before seeking desired risk
- prefer objective signals to discretionary opinions
- keep risk small enough that re-hedging and re-allocation remain possible
- use liquidity and execution quality as part of the edge, not as afterthoughts

Core Concepts

Statistical Arbitrage

Statistical arbitrage by cointegration trades the mean reversion of a stationary spread or residual built from two nonstationary price series.

Volatility Trading

Simple volatility trading uses delta hedging to isolate a long-volatility position. The trader seeks profits when:

- gamma gains offset theta decay
- vega gains add net profit because implied volatility rises

Gamma Scalping

Gamma scalping extends the volatility trade by minimizing or neutralizing vega. The portfolio aims to be:

- delta neutral
- vega neutral or near-neutral
- gamma positive
- theta negative

The trader then repeatedly monetizes price movement away from the last hedge point.

Vega Scalping

Vega scalping is an intraday long-volatility mean-reversion trade. It relies on intraday implied volatility behaving like a stationary series often enough that sharp downside dislocations in IV revert toward the intraday mean.

Shared Risk Rule

Never treat hedging as the strategy itself. Delta hedging is a prerequisite for volatility strategies, not the full trading edge.

Data Requirements

The system should not generate or approve trades unless the following data is available and current:

- underlying price history
- adjusted close history for stocks used in pair trading
- aligned business-day calendars
- option chain data with strikes, expiries, bid, ask, mid, volume, and open interest
- implied volatility
- Greeks or a reliable pricing engine
- earnings calendar
- corporate actions calendar
- borrow availability for short stock and short options where relevant
- margin estimate

Execution Requirements

The source material strongly implies algorithmic execution for options strategies. In this project, the minimum execution standard should be:

- spread-aware order construction
- synchronized multi-leg submission
- slippage caps
- rejection handling
- partial-fill handling
- ability to recompute hedge ratios after fill drift

Manual legging is acceptable only for observation or paper rehearsal, not as the target production workflow.

Retail Constraints To Respect

These strategies are institutionally inspired but must be filtered through retail reality:

- contract granularity makes perfect hedging difficult
- high-priced underlyings increase stock-hedge margin
- low-liquidity options can erase edge through spread costs
- short locates and borrow fees may invalidate otherwise attractive setups
- frequent hedging can turn a mathematically attractive trade into a net loser after costs

Shared Pre-Trade Checklist

No trade should be submitted unless all answers below are acceptable:

1. Is the signal objective and reproducible?

2. Is the instrument liquid enough to enter and exit cleanly?
3. Is the trade small enough to rebalance?
4. Are event risks known and intentional?
5. Does the strategy match the current regime instead of fighting it?
6. Are margin, borrow, and cost assumptions still valid now?
7. Is there a clear exit, stop, and time-based decommission rule?

Shared Kill Conditions

Abort or flatten if any of the following becomes true:

- liquidity collapses or spreads blow out beyond limits
- a hedge leg becomes unavailable
- the model input is stale or clearly corrupted
- an earnings or news event appears that the setup was not designed to absorb
- required neutrality cannot be restored within cost limits
- residual delta, vega, or spread risk exceeds portfolio limits
- the strategy's core assumption no longer holds

Strategy Selection Guide

Strategy	Best Use Case	Primary Edge	Main Risk	Typical Horizon
Statistical Arbitrage by Cointegration	Two related assets with stable economic linkage and stationary residual	Residual mean reversion	Relationship breakdown	Days to weeks
Simple Volatility Trading	Options look cheap versus forecast volatility	Long gamma plus long vega	IV does not rise, theta bleed	Same day to next day
Gamma Scalping	Need long gamma but want less IV risk, especially around gaps or high realized movement	Positive gamma with muted vega	Quiet market, term-structure distortion, execution complexity	Intraday to next day
Vega Scalping	Intraday IV drops below its mean in a liquid ATM option	IV mean reversion	IV keeps falling or loses stationarity	Intraday only

Shared Mathematical Reference

Linear Regression For Cointegration

- $\hat{Y} = mX + b$
- $e = Y - \hat{Y}$
- $m = [n * \sum(xy) - \sum(x)\sum(y)] / [n * \sum(x^2) - (\sum(x))^2]$
- $b = [\sum(y) - m * \sum(x)] / n$

Where:

- Y is the dependent asset
- x is the independent asset
- e is the residual or spread to be traded

Residual Normalization

Useful project metric:

- $z = (e - \text{mean}(e)) / \text{std}(e)$

The source text describes the same idea using mean and standard deviation bands.

Stationarity Intuition

If a series has:

- roughly stable mean
- roughly stable standard deviation

then mean-reversion logic can be used more safely. If it has a unit root or drifting mean, mean-reversion assumptions are unreliable.

Volatility Forecasting

The source volatility books use GARCH(1,1) as the main institutional forecast:

- $\sigma_n^2 = \gamma * \text{VL} + \alpha * u_{(n-1)}^2 + \beta * \sigma_{(n-1)}^2$

with source-example weights often described as:

- $\gamma = 5\%$
- $\alpha = 5\%$
- $\beta = 90\%$

Annualization:

- $\sigma_{\text{annual}} = \sigma_{\text{daily}} * \sqrt{252}$

Common Option Exposures

- Delta: directional sensitivity
- Gamma: rate of change of delta with underlying movement
- Theta: time decay
- Vega: sensitivity to implied volatility

Long options generally imply:

- positive gamma
- positive vega
- negative theta

Strategy 1: Statistical Arbitrage By Cointegration

Objective

Trade the mean reversion of a stationary residual extracted from two economically related but individually nonstationary assets.

Core Thesis

Two price series can each trend, gap, and change volatility while their residual still oscillates around a stable mean. The trade is not about forecasting either asset on its own. It is about exploiting dislocations in their relationship.

What Must Be True Before A Trade Exists

All of the following should be true:

1. The pair has a plausible economic relationship.
2. The regression residual is stationary enough to trade.
3. The current residual deviation is large enough to justify entry.
4. The expected time-to-mean-reversion is not too long.
5. There is no obvious fundamental reason the relationship should permanently break.

Best Pair Types

Highest-priority candidates:

- same sector, same country, same macro drivers
- same industry but different capitalization
- related ETFs and constituents
- assets with shared commodity, rate, or policy sensitivity

Lower-priority or higher-risk candidates:

- pairs joined only by statistical coincidence
- pairs with opposite business drivers unless the economic context clearly supports the trade direction
- pairs near mergers, restructurings, litigation, or earnings shocks

Signal Construction

Step 1: Build The Pair

- choose a candidate pair with economic logic
- align the price history
- prefer adjusted closing prices

Step 2: Run Regression

Model one asset as a function of the other:

- dependent asset: Y
- independent asset: X
- fitted line: $\hat{Y} = mX + b$
- residual: $e = Y - \hat{Y}$

Step 3: Test The Residual

The residual must be stationary or near-stationary by the system's acceptance thresholds. Useful source-aligned tests:

- Engle-Granger cointegration test
- Dickey-Fuller or Augmented Dickey-Fuller
- Phillips-Perron

Step 4: Measure Spread Deviation

Track:

- residual mean
- residual standard deviation
- z-score
- half-life estimate

Step 5: Evaluate Multi-Window Stability

The source material recommends evaluating multiple lookback windows rather than trusting one chart. Practical project windows:

- 20

- 40
- 60
- 80
- 100
- 120
- 140
- 160
- 180
- 200
- 220
- 250

Entry Rules

The source framework is:

- enter when the residual deviates beyond 2 standard deviations from the mean
- more selective traders may use 2.5 standard deviations
- avoid normal entries beyond 3 standard deviations because that typically indicates stress or breakdown risk rather than a standard reversion setup

Direction:

- residual too low: buy Y, sell X
- residual too high: sell Y, buy X

Exit Rules

Primary exit:

- close at residual mean reversion

Secondary exit:

- close when time in trade materially exceeds expected half-life
- close if the economic rationale changes before mean reversion occurs

Stop Rules

Source-aligned stop:

- stop around 3 standard deviations from the mean

Rationale:

- if the residual pushes through the 3-sigma area, the stationarity assumption may be failing or the pair may be repricing structurally

Position Sizing

The hedge ratio comes from the regression slope.

Example from the source material:

- $y = 11.736x + 25.33$
- buy 1 unit of Y
- sell approximately 11.736 units of X

Project implementation guidance:

- calculate slope-based hedge ratio
- then apply rounding and broker-lot constraints
- reject trades where rounding creates too much residual exposure
- cap gross exposure, pair exposure, and sector exposure

Time Management

The source material treats half-life as important operational information, not just a statistic.

Use half-life to:

- prioritize faster pairs
- size time-based stops
- rank capital efficiency

If a pair takes too long to revert, capital is trapped and opportunity cost rises.

Risk/Reward Framing

Source examples:

- entry at 2-sigma, stop at 3-sigma, target at mean: roughly 1:2
- entry at 2.5-sigma, stop at 3-sigma, target at mean: roughly 1:5

Do not over-trust these ratios mechanically. Realized fills, widening spreads, and fundamental shocks can worsen effective risk/reward materially.

Capital Allocation

Never allocate all capital to one pair.

Preferred project policy:

- equal-risk or capped-risk allocation across multiple valid pairs

- diversify non-systemic risk across names and sectors
- keep portfolio concentration low enough that one broken pair cannot dominate P/L

Scenario Guide

Scenario A: Same-Sector Mean Reversion

Example pattern:

- two banks diverge sharply
- residual hits entry band
- no major event separates one from the other

Action:

- this is the cleanest cointegration trade
- allow normal entry if stationarity and liquidity filters pass

Scenario B: Opposite Macro Sensitivity But Economically Consistent Trade

Example pattern:

- one asset benefits from the current environment
- the other is harmed by it
- the signal points in the same direction as macro logic

Action:

- possible trade, but treat as higher-complexity
- require explicit qualitative approval

Scenario C: Signal Conflicts With Economics

Example pattern:

- the model says buy a structurally damaged airline and short resilient big tech during crisis conditions

Action:

- reject the trade
- the econometric trigger is not sufficient on its own

Scenario D: Earnings Near One Leg

Action:

- reject or defer unless earnings risk is intentionally part of the design
- one-leg earnings can permanently distort the relationship

Scenario E: Spread Does Not Revert On Schedule

Action:

- reduce confidence
- review half-life breach
- close if capital is better deployed elsewhere

Scenario F: Residual Breaks Through Stop

Action:

- close decisively
- do not average down automatically
- flag the pair as potentially decommissioned pending re-test

Failure Modes

- spurious cointegration
- unstable hedge ratio
- one-window-only validity
- earnings or news shock on one leg
- regime change
- borrow cost or short restriction
- poor data alignment
- false confidence from test statistics

Best Practices For This Project

- rank pairs by both econometrics and economic context
- store rolling test history, not only current values
- log why a pair was approved, rejected, or decommissioned
- distinguish signal strength from execution feasibility

Strategy 2: Simple Volatility Trading

Objective

Trade volatility directly by creating a delta-neutral long-volatility position when implied volatility looks cheap relative to forecast volatility.

Core Thesis

Once delta is hedged away, the trader is no longer primarily trading direction. The position becomes a volatility trade that benefits when:

- realized movement generates gamma gains
- implied volatility rises and produces vega gains

The source books describe the ideal outcome as:

- gamma pays for theta
- vega delivers net profit

Minimum Conditions For A Valid Setup

1. The option chain is liquid.
2. The selected option is near-ATM.
3. Time to expiry is close enough to have meaningful gamma, but not so close that risk becomes chaotic.
4. Implied volatility is below the forecast from $GARCH(1, 1)$.
5. The trader can create and maintain a delta hedge.

Option Selection

Source-aligned selection rules:

- choose only high-liquidity options
- choose the most ATM strike
- prefer a close expiry, typically around 15 to 30 DTE
- avoid less than 10 DTE for routine use

Why this matters:

- ATM improves symmetry to both upside and downside movement
- near expiry increases gamma and theta while reducing vega
- too-close expiry makes Greeks more violent and execution less forgiving

Entry Signal

Primary source rule:

- enter when option implied volatility is below annualized $GARCH(1, 1)$ forecast volatility

This is a "cheap volatility" signal.

Position Construction

The source books allow two delta-hedging paths:

1. long calls + short underlying

2. long calls + long puts in the proportions required to neutralize delta

For options-only delta neutrality (same strike and expiration), define total option count N :

- number of **call** contracts = $N \times \text{put delta}$
- number of **put** contracts = $N \times \text{call delta}$

Project interpretation:

- use stock hedge if the stock is liquid and margin permits
- use options-only hedge when capital efficiency is needed and the desk accepts larger Greek leverage

Greek Profile To Seek

- delta approximately zero
- gamma positive
- vega positive
- theta negative

Trade Management

The key operational reference is the gamma-theta breakeven. The source material treats it as the price movement from the last hedge point needed for gamma gains to offset one day of theta decay. It is often described as being around 1%, but the project should calculate it from current Greeks rather than hard-code it.

Management rule:

- re-hedge when price moves away from the last hedge point by the gamma-theta breakeven

The re-hedge can be done by:

- increasing the hedge
- reducing the option position
- changing the mix of calls and puts

Exit Rules

Source-aligned default:

- hold $D+0$ or $D+1$

Carry to next day only if:

- gamma has already paid for theta at least once
- the setup still supports a long-volatility view
- there is no event likely to crush IV against the position

Stop Logic

Simple volatility trading does not use a single universal sigma stop like vega scalping. The trade should instead be stopped or avoided when:

- IV keeps falling after entry
- realized movement is too small to pay for theta
- delta cannot be maintained cheaply
- the original forecast advantage disappears

Scenario Guide

Scenario A: Normal Cheap-Vol Setup

Condition:

- $IV < \text{GARCH forecast}$
- liquid ATM option
- no extreme event distortion

Action:

- valid simple volatility trade candidate

Scenario B: Black Swan While Already Long Vol

Condition:

- major shock occurs after entry
- price gaps hard
- IV spikes

Action:

- take profits aggressively or re-hedge immediately
- do not assume the outsized move will persist in a tradable way

This is the most favorable path for the strategy.

Scenario C: Quiet Market After Entry

Condition:

- price stays near hedge point
- IV does not rise

Action:

- theta becomes the dominant force
- exit early rather than hoping for late rescue

Scenario D: Earnings Release Ahead

Condition:

- company earnings are imminent

Action:

- do not treat this as a routine simple volatility trade
- simple long-vol is vulnerable to post-event IV crush
- prefer gamma scalping if the goal is to capture gap risk with reduced vega exposure

Scenario E: High-Priced Underlying In Small Account

Condition:

- underlying price makes stock hedging capital-intensive

Action:

- prefer options-only delta hedge if execution quality remains acceptable
- otherwise reject the trade

Failure Modes

- IV does not rise despite the forecast
- GARCH is distorted after crisis periods
- theta bleed outpaces gamma
- liquidity is poor
- stock hedge consumes too much capital
- small contract count leaves large residual delta
- execution is too slow to preserve neutrality

Best Practices For This Project

- compute expected edge net of spread, slippage, commissions, and financing
- calculate whether one gamma-theta payment is realistically achievable
- downgrade or block signals in post-shock regimes
- do not hold for multiple days unless the trade repeatedly proves itself

Strategy 3: Gamma Scalping

Objective

Capture price movement through positive gamma while reducing or neutralizing exposure to changes in implied volatility.

Core Thesis

Simple volatility trading depends heavily on vega. Gamma scalping removes much of that dependency by combining short-dated and longer-dated options so that the portfolio remains:

- delta neutral
- vega neutral
- gamma positive
- theta negative

The trader then "scalps" the gamma effect as price moves away from the last hedge point.

Why Gamma Scalping Exists

This strategy is especially useful when:

- you want long convexity
- realized movement is likely
- implied volatility direction is uncertain
- an earnings gap is possible
- volatility is already high, making plain long-vega trades less attractive

Instrument Design

Source-aligned construction:

- buy short-dated calls
- short longer-dated calls in a quantity that neutralizes vega
- short stock to neutralize residual delta

Reason:

- shorter-dated options have more gamma and theta, less vega
- longer-dated options have more vega, less gamma and theta

By selling fewer or differently sized longer-dated options against the near-dated long options, the portfolio can neutralize vega while keeping net gamma positive.

Options-Only Variant

The source material also describes a four-leg options-only version:

- short-dated long calls
- longer-dated short calls
- short-dated long puts

- longer-dated short puts

Use case:

- lower stock-margin dependence
- higher structural complexity
- much stronger need for synchronized execution

Minimum Conditions For A Valid Setup

1. Both expiries are liquid.
2. Term structure is not obviously distorted against the trade.
3. Vega can be neutralized while keeping net gamma meaningfully positive.
4. Residual delta can be hedged within cost limits.
5. The expected realized movement is large enough to overcome theta.

Valid Entry Modes

The source books support three main reasons to open gamma scalping:

1. $IV < \text{GARCH forecast}$ and volatility appears cheap.
2. An earnings release is imminent and the goal is to capture the gap while muting IV-crush risk.
3. Volatility is already high and price is actively fluctuating, so realized movement may pay for theta even if IV later falls.

Entry Rules

Standard Entry

- choose same-strike short-dated and longer-dated options
- buy the shorter-dated call(s)
- short the longer-dated call(s) until vega is neutralized
- short stock to neutralize resulting delta

Earnings Entry

- open the trade one day before earnings
- the goal is the overnight gap, not a long holding period

Intraday Gamma Entry

- valid when price is already active and hedging costs are manageable
- intraday theta is less important than overnight theta

Greek Management

Greek neutrality is local, not permanent.

As price and time move:

- delta drifts
- vega neutrality drifts
- gamma and theta change

Therefore:

- re-hedge delta and vega whenever the move from the last hedge point reaches the gamma-theta breakeven

Exit Rules

Exit depends on why the trade was opened:

Cheap-Vol Or Standard Gamma Setup

- hold $D+0$ or $D+1$
- extend only if theta has been paid for and the structure remains favorable

Earnings Gap Setup

- close after the gap if the move delivered the intended gamma gain
- optionally extend for one more session only if realized movement remains high and the Greeks can be restored cheaply

Intraday Gamma Setup

- close same day unless the trade was intentionally structured for overnight exposure

Scenario Guide

Scenario A: Earnings Gap Down Or Up

Condition:

- stock gaps sharply after earnings
- IV may rise or collapse

Action:

- gamma is the target exposure
- re-check delta and vega immediately after the gap
- close or re-neutralize based on whether the move exceeded the gamma-theta threshold

This is the flagship gamma-scalping scenario.

Scenario B: High Volatility, Big Intraday Swings

Condition:

- realized movement is large
- term-structure hedge remains stable

Action:

- re-hedge repeatedly at rule-based intervals
- lock in gamma gains as price oscillates

Scenario C: Quiet Market

Condition:

- price stays near hedge point

Action:

- theta slowly dominates
- close rather than forcing the trade

Scenario D: IV Term-Structure Distortion

Condition:

- short-dated IV is abnormally rich while longer-dated IV is abnormally cheap

Action:

- reject or reduce the trade
- the book explicitly warns that term-structure correction can create losses even if vega is neutral at entry

Scenario E: Post-Gap Greek Drift

Condition:

- large move happens
- delta and vega neutrality disappear

Action:

- do not assume the original hedge still exists
- either re-solve the structure or flatten

Failure Modes

- not enough movement to beat theta

- poor liquidity in longer-dated legs
- imperfect vega hedge from coarse contract sizing
- post-shock GARCH distortion
- partial fills across three or four legs
- term-structure misalignment
- overconfidence in the "vega neutral" label after a large move

Best Practices For This Project

- solve quantities numerically instead of using rough intuition
- store hedge point and rebalance threshold explicitly
- support separate entry modes: `cheap_vol_mode`, `earnings_gap_mode`, `high_realized_vol_mode`
- require stronger execution checks than for simple volatility trades

Strategy 4: Vega Scalping

Objective

Capture intraday mean reversion in implied volatility through a delta-neutral long-volatility structure.

Core Thesis

The source book argues that although implied volatility may trend over multi-day horizons, it is often stationary enough intraday to revert toward its mean. Vega scalping exploits this by entering only after a downside dislocation in intraday IV and exiting on mean reversion.

Defining Characteristics

- intraday only
- long volatility only
- delta neutral at entry
- gamma positive as a beneficial secondary exposure
- theta largely ignored because the trade should not be held overnight

Minimum Conditions For A Valid Setup

1. The selected ATM option is highly liquid.
2. Intraday IV has a meaningful and stable enough mean.
3. Intraday standard deviation is not exploding in a way that invalidates the signal.
4. A delta-neutral hedge can be created immediately.
5. The trade can be flattened before the end of the session.

Option Selection

Source-aligned rules:

- choose an ATM option
- find the best compromise between longer expiry and strong liquidity
- avoid ultra-near expiry when Greek instability becomes too high

Entry Rules

Primary source rule:

- enter only when intraday implied volatility is 2 standard deviations below its intraday mean

Do not invert the rule.

The source material is explicit:

- never treat 2-sigma above mean as a short-volatility entry

Position Construction

Two possible structures:

1. long option(s) plus stock hedge
2. calls and puts combined to neutralize delta

Project preference:

- choose the construction with the better combined score for liquidity, slippage, and margin

Exit Rules

Primary exit:

- close when IV returns to its intraday mean

Stop:

- stop at 3 or 4 standard deviations below mean, depending on configured risk tolerance

Time exit:

- flatten same day, always

Why The Strategy Can Work

The source rationale is:

- a stationary series oscillates around a fixed mean
- if IV drops far enough below the mean, it tends to revert

- because the trade is long volatility, unexpected market agitation after entry helps rather than hurts

Scenario Guide

Scenario A: Clean Intraday IV Flush

Condition:

- liquid ATM option
- IV prints at least 2-sigma below mean
- spreads remain tight

Action:

- valid vega-scalp entry
- exit at mean reversion

Scenario B: News Shock After Entry

Condition:

- sudden macro or company-specific news breaks
- IV spikes

Action:

- favorable outcome
- take gains rather than waiting for a perfect textbook exit

Scenario C: Quiet Tape

Condition:

- market remains still
- IV continues to drift lower

Action:

- stop out at configured sigma threshold
- do not widen the stop merely because the trade was "supposed" to revert

Scenario D: Intraday Stationarity Breakdown

Condition:

- intraday IV mean is present but standard deviation becomes unstable

Action:

- downgrade or block new entries

- active positions should respect hard stop and same-day flattening

Scenario E: Illiquid Chain

Condition:

- wide spreads or thin order book

Action:

- reject the setup even if the IV signal is strong

Failure Modes

- IV continues to fall
- variable intraday variance creates moving-goalpost behavior
- stale IV measurements
- execution latency destroys the entry edge
- residual delta remains because of coarse contract sizing
- the chain is liquid enough to enter but not liquid enough to exit cheaply

Best Practices For This Project

- build contract-specific intraday IV history
- compute rolling mean and rolling standard deviation with outlier handling
- require same-day flattening
- log realized vega P/L separately from gamma P/L

Cross-Strategy Scenario Map

When Volatility Looks Cheap Versus Forecast

Preferred order:

1. simple volatility trade
2. gamma scalping if IV direction is uncertain or event risk is present

When A Large Earnings Gap Is Likely

Preferred order:

1. gamma scalping
2. avoid simple volatility trade through the event unless explicitly justified
3. avoid cointegration if only one leg has earnings risk

When Intraday IV Flushes Hard In A Liquid ATM Option

Preferred order:

1. vega scalping

When Realized Movement Is High But IV Is Already Elevated

Preferred order:

1. gamma scalping

Reason:

- it reduces the problem of paying rich vega while still keeping long gamma

When Economic Context Is Strong But Options Liquidity Is Weak

Preferred order:

1. cointegration

When The Market Is Post-Shock And Models Are Distorted

Preferred order:

- reduce or block all model-driven volatility trades until normalization
- cointegration can also be degraded if relationships have structurally changed

Portfolio-Level Rules

Diversification

Do not let one strategy dominate all risk by accident. The system should monitor:

- gross exposure
- net exposure
- sector concentration
- earnings concentration
- aggregate delta
- aggregate gamma
- aggregate vega
- aggregate theta

Capital Allocation Policy

Suggested hierarchy:

1. reserve capital for hedging and forced exits first
2. allocate only a fraction of capital to any one pair or volatility structure
3. reduce size when liquidity is thin or hedging precision is poor
4. reduce size after major shocks until models re-stabilize

Cost Awareness

Every signal should be scored after:

- commissions
- bid/ask spread
- slippage
- borrow costs
- financing
- expected re-hedge count

The project should reject any trade whose theoretical edge disappears after realistic costs.

Supervised Execution Runbook

Pre-Approval Packet

Before operator approval, the system should present:

- strategy type
- instrument(s)
- market condition summary
- entry rationale
- hedge construction
- size and margin estimate
- stop, target, and time exit
- known event risks
- reasons the trade could fail

Post-Entry Automation

Once approved and filled, the system may automate:

- delta-maintenance hedges
- risk-reduction exits
- stop logic

- same-day flattening for vega scalping
- alerts for lost neutrality or abnormal costs

Human Escalation Triggers

Require operator review if:

- a leg fails to fill
- the hedge cannot be restored
- slippage exceeds configured maximum
- a major event occurs mid-trade
- the system recommends holding beyond the default horizon

Implementation Notes For The Project

Strategy Modules To Support

The analytics and execution stack should maintain dedicated logic for:

- cointegration and residual analytics
- volatility forecasting
- option selection
- Greeks aggregation
- hedge solving
- scenario tagging
- execution orchestration
- post-trade attribution

Key Metrics To Persist

At minimum, persist:

- signal timestamp
- instrument identifiers
- lookback window(s)
- hedge ratio
- residual z-score
- ADF and Engle-Granger statistics
- half-life
- IV, forecast volatility, and IV z-score
- delta, gamma, vega, theta at entry and after each rebalance

- realized and unrealized P/L
- slippage and spread capture
- reason for exit

Conservative Interpretation Of Source Claims

The source PDFs are valuable and operationally rich, but the project should interpret some claims conservatively:

- statistical confidence from ADF or cointegration tests is not the same as realized win rate
- "volatility is cheap" is a probabilistic statement, not a guarantee
- "vega neutral" is local and can disappear quickly after a large move
- an elegant Greek structure can still fail because of liquidity, event risk, or execution quality

Final Operating Rules

1. Trade only when the setup matches the strategy's intended regime.
2. Do not force a strategy into the wrong market condition.
3. Respect liquidity, margin, and event-risk filters before respecting model beauty.
4. For volatility strategies, hedge quality is part of the trade thesis.
5. For cointegration, economics decides whether the statistically valid trade should exist.
6. Do not widen stops to rescue a broken thesis.
7. Flatten when the core assumption fails.
8. Review every completed trade to refine scenario filters, cost assumptions, and execution rules.

Complete Source Tables And Interpretations

This section preserves every execution-critical table and rule list from the four source PDFs, with plain-language interpretation for the trading bot. Tables that appear as charts or screenshots in the originals are reconstructed here as structured reference data.

Table Index By Source Document

Source Document	Table / Rule Set	Primary Use For The Bot
Statistical Arbitrage by Cointegration	Empirical rule bands, entry/exit/stop framework, platform criteria, AMZN/FB example	Pair signal validation, sizing, and P/L attribution
Volatility Trading	Option selection rules, rules 1–10, GARCH example, Company Z walkthrough	Cheap-vol entry, delta hedge, gamma-theta management

Gamma Scalping	Greeks-vs-time, three entry modes, Intel earnings example, rules 1–10	Vega-neutral construction, earnings gap, intraday scalp
Vega Scalping	Seven intraday rules, IV mean-reversion example	Intraday IV z-score entry and same-day exit

Statistical Arbitrage By Cointegration — Source Tables

Table SA-1: Empirical Rule For Residual Bands

Standard Deviation Band	Approximate Share Of Observations	Bot Interpretation
Within $\pm 1\sigma$ of residual mean	68%	Normal oscillation; no entry
Beyond $\pm 2\sigma$ of residual mean	~5% tail beyond band (95% inside)	Primary entry trigger when economics agree
Beyond $\pm 3\sigma$ of residual mean	~0.3% tail beyond band (99.7% inside)	Stop zone; stationarity assumption may be failing

Interpretation: The source treats 2σ as the standard mean-reversion entry and 3σ as the statistical stop. Entries beyond 3σ are discouraged because they usually indicate stress or structural repricing, not a routine reversion setup.

Table SA-2: Entry, Exit, And Stop Framework

Parameter	Source Rule	Bot Action
Entry (standard)	Residual escapes 2 standard deviations (2σ) below/above mean	Open spread: buy Y / sell X if residual low; reverse if high
Entry (selective)	Residual escapes 2.5 standard deviations (2.5σ)	Same direction; higher selectivity, better nominal R:R
Entry (avoid)	Beyond 3 standard deviations (3σ)	Block new entry unless explicitly flagged as stress research
Target	Residual mean	Primary take-profit
Stop	3 standard deviations (3σ) from mean	Hard stop; flag pair for re-validation
Risk/reward at 2σ entry	~1:2 (2 standard deviation entry → mean target, 3 standard deviation stop)	Size assuming stop is farther than target

Risk/reward at 2.5σ entry	~1:5	Prefer when half-life and liquidity still support hold
Time stop	Half-life from Ornstein-Uhlenbeck	Close or downgrade if reversion exceeds expected half-life

Interpretation: Entry uses **standard deviation** bands around the residual mean (see Table SA-1). **Stop loss** at **3 standard deviations** from the mean. Econometrics triggers timing; economics decides whether the trade should exist at all. Never auto-approve on z-score alone.

Table SA-3: Multi-Window Cointegration Lookback

Lookback Window (Business Days)	Bot Use
20, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 250	Test cointegration separately in each window

Interpretation: A pair cointegrated in only one window is weaker than a pair cointegrated across many windows. Prefer pairs with (a) more cointegrated windows, and (b) simultaneous entry signals in multiple windows.

Table SA-4: Pairs Platform Selection Criteria

#	Criterion	Minimum Standard	Bot Interpretation
1	Number of cointegrated periods	Higher is better	Rank pairs by count of valid windows
2	Simultaneous entry signals	Multiple windows at entry band	Stronger signal when several windows agree
3	ADF stationarity	95% or 99% confidence	Required filter; log confidence at signal time
4	Engle-Granger cointegration	95% or 99% confidence	Required filter alongside ADF
5	Half-life (Ornstein-Uhlenbeck)	Shorter is better	Use for time stop and capital ranking
6	Economic context	Must support direction	Human or rules engine must confirm; block conflicting macro logic

Interpretation: ADF/Engle-Granger confidence is a model statement about stationarity, not a realized win rate. The bot should still enforce economics, earnings, and liquidity filters.

Table SA-5: Linear Regression And Hedge Ratio (Amazon / Facebook Example)

Field	Value	Interpretation
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Regression equation	$Y = 11.736X + 25.33$	Y = Amazon (dependent), X = Facebook (independent)
Hedge ratio (slope)	11.736 FB shares per 1 AMZN share	Size short leg as slope $\times$ long shares, then apply lot rounding
Entry trigger	Residual below -2σ	Buy AMZN, sell FB
Hold period (example)	24 business days	Compare to half-life estimate

Table SA-6: AMZN / FB Entry And Exit Prices

Leg	Entry Price	Exit Price	Direction In Example
Amazon (Y)	\$2,436.88	\$2,692.87	Long (+10.5%)
Facebook (X)	\$234.91	\$216.08	Short (+8.01% on short leg)

Interpretation: Both legs contributed profit because the spread reverted while each leg moved favorably relative to its side of the trade.

Table SA-7: AMZN / FB Position, Margin, And P/L Breakdown

Field	AMZN (Long)	FB (Short)	Notes
Shares / contracts at entry	100	1,176	Rounding (rounded from 11.736 to 1,176)
Entry notional / margin	\$243,688.00	\$275,416	FB, 25% PL assumes 100% short margin
Total entry margin	—	\$51,934.00	Cap reference for retail sizing
Long-side P/L	\$25,599.00	—	
Short-side P/L	—	\$22,144.08	
Total P/L	\$47,743.08		

Return on margin	9.18%		Over 24 business days
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Interpretation: Reject or downsize trades where rounding error on the hedge ratio creates material residual exposure. Log realized vs expected hedge ratio at fill.

Table SA-8: Arbitrage Scenario Matrix

Scenario	Economic Relationship	Signal Aligns With Macro?	Bot Disposition
Same-sector mean reversion	Same macro drivers	Yes or neutral	Allow if tests pass
Opposite macro sensitivity	One helped, one hurt	Yes, if buying winner / selling loser	Allow with higher scrutiny
Opposite macro sensitivity	One helped, one hurt	No (buy loser / sell winner)	Reject
Earnings on one leg	Any	N/A	Defer unless earnings are intentional
COVID-style sector divergence	Statistically cointegrated	No	Reject despite z-score

Volatility Trading — Source Tables

Table VT-1: Option Selection Rules

#	Rule	Bot Interpretation
1	Trade only high-liquidity options	Block illiquid chains; enforce spread cap
2	Choose ATM strike	Maximizes gamma symmetry and Greek magnitude
3	Choose near expiry (~15–30 DTE)	More gamma/theta, less vega
4	Avoid <10 DTE routinely	Black-Scholes distortions; extreme Greek risk

Table VT-2: Core Volatility Trading Rules (Source Numbering)

#	Rule	Bot Interpretation
3	Enter when option implied volatility (IV) is below the GARCH(1,1) forecast	Primary cheap-vol signal; IV must be lower than GARCH
4	Delta hedging is prerequisite, not the strategy	Must neutralize delta before managing gamma/vega/theta
5	Always long options (positive gamma); never short vol for this strategy	Long calls or long call+put structures only
6	Delta hedge via (a) long calls + short stock, or (b) long calls + long puts same strike/expiry	Choose path by margin and liquidity score
7	Stock hedge: buy X calls $\approx$ long X $\times$ call delta shares; hedge by shorting that many shares	Default institutional path when stock is liquid
8	Options-only hedge: total contracts N; calls = N $\times$ put delta ; puts = N $\times$ call delta	Lower margin; stronger Greek leverage
9	Re-hedge when price moves away from last hedge point by gamma-theta breakeven (~1%)	Automated rebalance trigger
10	Re-hedge by increasing or decreasing position	Track realized P/L and floating P/L separately

Table VT-3: GARCH(1,1) Parameter Reference

Parameter	Typical Weight	Role
γ (gamma)	5%	Weight on long-run variance VL
α (alpha)	5%	Weight on prior squared return
β (beta)	90%	Weight on prior variance estimate
Constraint	$\gamma + \alpha + \beta = 100\%$	Must hold in implementation

Formula: $\sigma_n^2 = \gamma \cdot VL + \alpha \cdot u_{(n-1)}^2 + \beta \cdot \sigma_{(n-1)}^2$

Table VT-4: GARCH Worked Example (Source Numbers)

Step	Calculation	Result
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Long-run variance VL	Sample variance of log returns	0.030753%
Prior squared return $u_{(n-1)^2}$	From series	0.036051%
Prior variance $\sigma_{(n-1)^2}$	From series	0.036595%
Today's variance σ_n^2	$5\% \cdot VL + 5\% \cdot u^2 + 90\% \cdot \sigma_{(n-1)^2}$	0.0362757%
Daily volatility σ_n	$\sqrt{(\text{variance})}$	1.9046%
Annualized volatility	$1.9046\% \times \sqrt{252}$	30.23%

Interpretation: Compare annualized GARCH forecast to annualized option IV. IV below forecast = cheap vol entry candidate. After black swan events, GARCH may be distorted — block or downgrade signals until normalization.

Table VT-5: Gamma-Theta Breakeven Management

Concept	Typical Value	Bot Interpretation
Gamma-theta breakeven	~0.96%–1% move from last hedge point	Distance required for one day's gamma to offset one day's theta
Half breakeven tactic	Re-hedge at half distance twice	Alternative when full breakeven is hard to reach
Carry permission	Paid breakeven $\geq$ once	May carry D+0 $\rightarrow$ D+1
Default horizon	D+0 or D+1	Do not carry multi-day without repeated theta payment

Table VT-6: Company Z Walkthrough — Small Account Delta Hedge

Step	Action	Result
Setup	Stock ~\$20; 10 ATM call contracts; 20 total option slots	Initial portfolio delta –92.56 (not neutral)
Hedge attempt A	Adjust puts to 8 contracts	Best delta +16.47 (too long)
Hedge attempt B	Adjust calls (preferred in example)	Delta –2.04; margin ~\$1,190
Price move	+1% to \$20.20 (breakeven distance)	Floating P/L +\$12.39 (~1% on margin); delta +125.82
Re-hedge	Sell 2 call contracts (reduce size)	Realized +\$19.26; floating –\$6.88; total unchanged
Overnight	–2% gap; IV 20% $\rightarrow$ 22%	Floating +\$120.40

Total trade	Realized + floating	\$139.66 profit on ~\$1,200 margin (~11.6%)
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Interpretation: Small contract counts prevent perfect delta neutrality. Bot must log residual delta, compare call vs put adjustment paths, and track realized vs floating P/L on every rebalance.

Table VT-7: Greek Profile Target — Simple Volatility Trade

Greek	Target Sign	Source Rationale
Delta	≈ 0	Remove directional exposure
Gamma	Positive	Gain from movement away from hedge point
Vega	Positive	Profit when IV rises toward GARCH forecast
Theta	Negative	Cost of carry; must be paid via gamma scalps

Table VT-8: Volatility Trade Exit Logic

Condition	Action
Gamma-theta breakeven paid $\geq$ once and thesis intact	May hold to D+1
IV rises (vega gains) + gap/move	Strong close candidate
IV falls and price quiet	Exit early; vega losses dominate
IV cheap vs GARCH at entry	Expect IV rise; if IV falls 3+ points anyway

Gamma Scalping — Source Tables

Table GS-1: Greeks Versus Time To Expiration

Greek	Relationship To Expiry	Near-Dated Option	Far-Dated Option
Gamma	Inversely proportional	Higher	Lower
Theta	Inversely proportional	Higher (more decay)	Lower
Vega	Directly proportional	Lower	Higher

Interpretation: Buy short-dated calls for gamma/theta; sell fewer long-dated calls to neutralize vega; short stock to neutralize delta. Result: delta neutral, vega neutral, gamma positive, theta negative.

Table GS-2: Target Portfolio Profile

Greek	Target	After Large Move
Delta	0	Drifts; must re-hedge
Vega	0	Drifts; must re-hedge
Gamma	Positive	Still positive but changes magnitude
Theta	Negative	Overnight theta matters for D+1 holds

Table GS-3: When To Perform Gamma Scalping (Three Source Modes)

Mode	Entry Condition	Primary Goal	Default Horizon
1 — Cheap vol	$IV < \text{GARCH}(1,1)$	Gamma pays theta; optional vega lift	D+0 / D+1
2 — Earnings gap	Open 1 day before earnings	Capture gap gamma; mute IV crush via vega hedge	Close after gap or next session
3 — High realized vol	IV already elevated; price active	Scalp gamma without betting on IV rise	Intraday or D+1

Table GS-4: Gamma Scalping Construction Steps

Step	Action	Bot Note
1	Select same-strike short-dated and long-dated calls	Prefer calls over puts (positive delta → short stock uses less margin)
2	Buy short-dated calls	Provides gamma and theta
3	Short long-dated calls until portfolio vega ≈ 0	Creates small negative gamma; net gamma stays positive
4	Short underlying to neutralize delta	Stock hedge path
Alt	Four-leg options-only	Short-dated long calls + long-dated short calls + short-dated long puts + long-dated short puts; requires synchronized algo

Table GS-5: Intel Earnings Gap Example (Source)

Field	Value	Interpretation
Pre-earnings price	\$60.50	
Post-gap price	\$52.14	–13.8% gap
IV change (simulation)	35% → 45%	Vega-neutral design limits IV path dependency
P/L with stock delta hedge	\$6,635	Gamma-driven
P/L options-only structure	\$13,272 (after theta)	Higher complexity, higher outcome in example
Post-gap Greeks	No longer delta/vega neutral	Mandatory re-solve or flatten

Table GS-6: Gamma Scalping Management Rules

Situation	Re-hedge Rule
Intraday gamma scalp	Re-neutralize delta and vega at each gamma-theta breakeven move
Overnight / earnings gap	Re-neutralize after gap if move $\geq$ gamma-theta breakeven
Intraday only	Theta less relevant same session; focus on gamma extraction
Term-structure distortion	Block entry if short-dated IV > long-dated IV (local max/min mismatch)

Table GS-7: Gamma Scalping Exit Rules

Opening Mode	Exit Rule
Cheap vol / standard	D+0 or D+1; extend only if breakeven paid and structure restorable

Earnings	Close after gap profit or extend one session if movement stays high
High realized vol intraday	Flatten same day unless intentionally structured overnight

Table GS-8: IV Term-Structure Risk (Hidden Loss Source)

Term Structure State	Risk	Bot Action
Short-expiry IV < long-expiry IV	Favorable	Preferred entry geometry
Short-expiry IV locally high; long-expiry locally low	Mean-reversion of IV spread can hurt	Reject or reduce size
Post-entry 20% IV drop (vega-neutral book)	Minimal P/L impact	Validates vega hedge purpose

Vega Scalping — Source Tables

Table VS-1: Stationarity Comparison For IV

Series Type	Mean	Std Dev	Predictable By Mean Reversion?	Bot Use
Multi-day IV	Often trends	Variable	Poor intraday	Use GARCH cheap-vol frame (Vol Trading)
Intraday IV	Approximately fixed	Mostly stable	Yes, when stable	Vega scalping universe
Intraday IV with variable variance	Fixed mean	Variable σ	Degraded	Downgrade signals; tighten stops

Interpretation: Vega scalping applies cointegration-style logic to intraday IV, not to price direction.

Table VS-2: Seven Vega Scalping Rules (Source)

#	Rule	Bot Interpretation
1	Choose ATM option	Maximizes vega/gamma tradeoff
2	Balance longer expiry vs liquidity	Prefer liquid chain over maximum vega

3	Monitor intraday IV series	Build contract-specific IV history
4	Enter when intraday IV is 2σ below intraday mean	Long vol only; delta hedge immediately
5	Never short vol when IV is 2σ above mean	Negative gamma; explicitly forbidden
6	Exit when IV returns to intraday mean	Primary take-profit
7	Stop at 3σ or 4σ below mean	Configurable by risk tolerance

Table VS-3: Vega Scalping Greek Focus (Intraday)

Greek	Relevance Intraday	Notes
Delta	Neutralize at entry	Prerequisite, not the edge
Gamma	Secondary benefit	Helps if price moves after entry
Vega	Primary edge	Mean reversion of IV
Theta	Largely ignored	D+0 only; flatten same session

Table VS-4: Vega Scalping Worked Example (Source)

Field	Value	Interpretation
Entry trigger	IV crosses -2σ below intraday mean	Long vol via delta-neutral calls+puts
IV distance to mean	~2 percentage points	Proxy for expected vega capture
Position margin	\$16,533	Size reference
Profit at mean reversion	\$1,652	~10% return; excludes possible gamma add-on
Leverage note	Options-only vs stock hedge	Stock hedge lowers leverage and risk

Table VS-5: Vega Scalping Failure And Antifragile Scenarios

Scenario	P/L Effect	Bot Action
IV reverts to mean	Profit	Standard exit

Breaking news spikes IV	Profit (long vega)	Take profit; do not wait for perfect mean touch
Quiet tape; IV drifts lower	Loss	Stop at 3σ – 4σ ; same-day flatten
IV variance unstable	Signal degradation	Block new entries

Shared Volatility Implementation Tables (All Options Strategies)

Table SH-1: Numbered Practical Aspects (Gamma / Vol Books)

#	Topic	Bot Requirement
1	Stock price and IV affect margin	Pre-trade margin model uses spot and IV
2	Small contract counts	Expect imperfect delta/vega neutrality; enforce minimum size or reject
3	Algorithmic execution	Multi-leg synchronized submit; no manual legging in production
4	GARCH after black swans	Block cheap-vol signals until model normalizes
5	Gamma/theta vs vega vs time	Use Table GS-1 for expiry selection
6	Avoid <10 DTE	Hard filter except research mode
7	Liquidity	Spread and OI filters; critical for long-dated gamma legs
8	Re-hedge by increasing or decreasing	Preserve realized + floating P/L ledger
9	Greek equilibrium is local	Recompute Greeks after every material move
10	Goal Seek / numerical hedge solve	Solve contract counts numerically; do not guess ratios

Table SH-2: Delta Hedge Construction Comparison

Method	Legs	Margin	Liquidity	Neutralization Quality
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Long calls + short stock	2	Higher if stock expensive	Stock + option	Usually best for whole shares
Long calls + long puts	2+	Lower	Option-only	Sensitive to contract granularity
Gamma: calls spread + stock	3+	Medium–high	Needs long-dated liquidity	Must solve vega and delta jointly
Gamma: four-option box	4	Lower stock margin	Hardest execution	Requires algo

Table SH-3: Horizon And Carry Matrix

Strategy	Allowed Hold	Carry Condition	Must Flatten By
Statistical arbitrage	Days–weeks	Half-life not breached	Stop, time stop, or mean hit
Simple volatility	D+0 or D+1	Gamma-theta breakeven paid	End D+1 unless re-approved
Gamma scalping	D+0, D+1, or post-earnings	Mode-dependent	Earnings mode: after gap unless extended
Vega scalping	Intraday only	N/A	Session close always

Table SH-4: Cross-Strategy Decision Matrix (Source Logic)

Market Condition	First Choice	Second Choice	Avoid
IV < GARCH; normal regime	Simple vol	Gamma scalping if IV path uncertain	Short vol
Earnings gap expected	Gamma scalping	—	Plain long vega through event
IV high; large realized moves	Gamma scalping	—	Simple long vega
Intraday IV -2σ vs intraday mean	Vega scalping	—	Short vol at $+2\sigma$
Stable pair; residual $\pm 2\sigma$; economics OK	Cointegration	—	Forced vol trade
Post-shock; GARCH distorted	Reduce all vol strategies	—	Blind GARCH entries

Bot Ingestion Notes For This Section

When indexing this document for RAG or strategy engines:

- Tag each table with strategy, topic, and source_doc metadata (for example: doc-gamma, gamma-theta-breakeven).
- Prefer retrieving entire table + interpretation chunk together; rows split across chunks lose execution context.
- Numeric examples (AMZN/FB, Company Z, Intel, Vega example) are calibration references, not guaranteed future performance.
- Economics gate (Table SA-8) overrides purely statistical triggers for cointegration.
- Rule 5 in Vega Scalping (never short vol at $+2\sigma$) is a hard constraint, not a suggestion.

Document Status

This file is intended to be the canonical high-level trade-execution reference for the project. The section **Complete Source Tables And Interpretations** consolidates all execution-critical tables and numbered rules from the four source PDFs. It should be updated whenever:

- a strategy rule changes
- new scenario handling is added
- broker or margin constraints change
- execution infrastructure improves
- backtest evidence invalidates a source-derived heuristic